MIL-F-50022 (CmlC)
23 June 1959
(See section 6)

MILITARY SPECIFICATION

FACEBLANK (FOR CD V-805 MASK)

1. SCOPE

1.1 Scope. This specification covers one type and six sizes of protective mask faceblanks.

1.2 Classification. Faceblanks shall be of the following sizes (see 3.2 and 6.2):

Size 1
Size 2
Size 3
Size 4
Size 5
Size 6

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-406 - Plastics, Organic: General Specification, Test Methods.

FSC 4240

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STANDARDS

FEDERAL

Federal Test Method Std. No. 601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-177 - Rubber Products, Terms for Visible Defects.

DRAWINGS

CHEMICAL CORPS

E5-2-908 - Faceblank Size 6.
E5-2-909 - Faceblank Size 5.
E5-2-910 - Faceblank Size 4.
E5-2-911 - Faceblank Size 3.
E5-2-912 - Faceblank Size 2.
E5-2-913 - Faceblank Size 1.

PUBLICATIONS

CHEMICAL CORPS PURCHASE DESCRIPTIONS

197-54-653 - Filter Material, Gas Aerosol, El7.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY OF TESTING MATERIALS

A.S.T.M. Standards 1955

Method D746

(Copies may be obtained from the American Society of Testing Materials, 1916 Race St., Philadelphia 3, Pa.)

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CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Molding plastic compounds. All molding compounds furnished under this specification shall be of a brand (manufacturer's proprietary designation for a special compound). New or reclaimed material of various brands shall not be mixed together.

3.1.1.1 Sun checking retardant. The plastic shall be formulated with a sun checking retardant.

3.1.1.2 Color. The color of the plastic shall be specified by contract or order (see 6.2).

3.1.2 Filter material. The material in the filter element shall conform to Purchase Description 197-54-653, type II.

3.2 Manufacturing. The faceblanks shall be manufactured in molds approved by the Government in accordance with the applicable drawings, as follows:

Size 1 - E5-2-913
Size 2 - E5-2-912
Size 3 - E5-2-911
Size 4 - E5-2-910
Size 5 - E5-2-909
Size 6 - E5-2-908

3.2.1 Molding compound. Test slabs, 6 by 6 by 0.075 ± 0.01 inches, shall be injection molded under the same conditions of time, temperature, and pressure and with stock of the same lot of molding compound used for the preproduction sample and regular production lot (see 3.3).

3.3 Preproduction sample. Prior to the start of regular production, a preproduction sample of 25 faceblanks, 55 test slabs and 10 molded samples (see 3.7) produced using the same methods, materials, and type of equipment as will be used during regular production. Any change in formulation, operating conditions, or molds shall require a new preproduction sample. Samples resulting from mold repair will not require a toxicity and charcoal poisoning tests.

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3.4 Physical properties.

3.4.1 Test slabs. The test slabs shall conform to the physical properties, before and after aging, as specified in table I.

Table I. Physical requirements

Property	Requirement		Test method
	before aging	after aging	
Tensile strength, p.s.i., minimum (min.)	1275	1275	4.6.1
Elongation, percent at breaking, min.	425	425	4.6.1
Modulous at 100 percent elongation	375 to 425		4.6.1
Compression set, at maximum (max.) load, percent max.	84		4.6.1
Hardness, Dura A	62 to 68	60 to 65	4.6.1
Tear resistance, pounds per inch, min.	130		4.6.1
Resiliant rebound, instrument reading	45 to 55		4.6.1.1
Flammability	Self extinguishing		4.6.1
Low temperature flexibility	pass at -40°F.		4.6.1.2
Specific gravity	1.23 to 1.27		4.6.1

3.4.2 Faceblanks. Specimens cut from the faceblank shall be tested for the hardness and the specific gravity requirements of paragraph 3.4.1.

3.5 Toxicity. The faceblank material shall be nontoxic to the skin when applied locally when tested as specified in 4.6.4. Thirty test slabs (of those in 3.3) shall be submitted for toxicity tests. The thirty test slabs shall be molded with the ranges of temperatures, pressures and cure times that the supplier will use during regular production. Samples shall be forwarded to the agency designated by the contracting officer. The samples shall be

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marked with the following information:

Sample for toxicity tests
Name of material
Manufacturer's designation
Manufacturer's name
Range of conditions
  Cure Temperatures:
  Cure Pressures:
  Cure Times:

The supplier shall make no change in plastic formulation or in the molding conditions submitted after receiving approval of a formulation and molding conditions from the procuring activity.

3.6 Latex adhesion properties. When tested as specified in 4.6.5 the average force of not less than 25 pounds per inch and no individual value of less than 20 pounds per inch shall be required to cause any separation at the interface between the filter material and the faceblank material.

3.7 Charcoal poisoning. The latex and faceblank material shall not poison the charcoal in the filter material. Prior to the start of regular production, molded samples (see figure 1) from the preproduction run shall be submitted for test of charcoal poisoning. The samples shall be tested as specified in 4.6.6. After approval of latex and molding compound formulations no change shall be made in the formulation of either the latex or molding compound without prior approval from the Government. (See 6.6)

3.8 Workmanship. The inner and outer surfaces of the faceblank shall be free from pits, lumps, porous areas, cracks, checks, flash, abrasions, blisters, discoloration, weld lines, burned spots, dull streaks, flow lines, shrinkage marks, sinks, orange peels or discoloration. The faceblank shall be free from contamination (foreign material) and damage such as tears, cuts or punctures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Supplier's responsibility. The supplier is responsible for the examination and testing as prescribed herein except for those examinations and tests reserved for performance by the Government.

4.1.2 Government's responsibility. The Government will accomplish the examinations set forth in 4.4.1 and tests set forth in 4.4.2.

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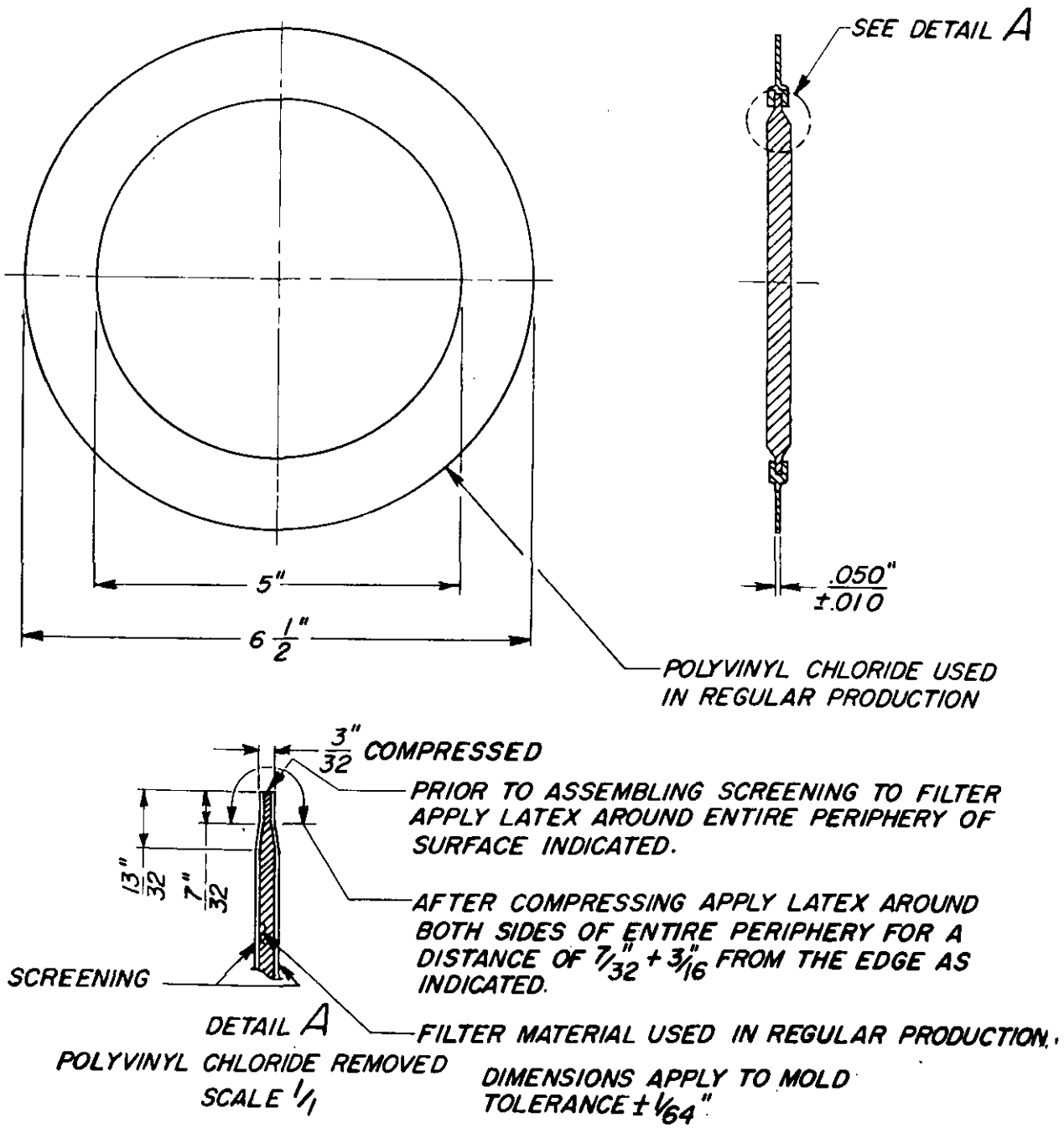


Figure 1 – Samples for test of charcoal poisoning

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4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all material and components are in accordance with paragraph 3.1 and section 5(see 6.3).

4.3 Alternate inspection(including testing) procedures. The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Preproduction sample inspection.

4.4.1 Examination. Samples shall be inspected for all requirements of the drawings and specifications.

4.4.2 Tests. Five (5) faceblanks selected at random shall be tested in accordance with paragraph 4.6.3 and 10 molded samples (see 3.7) shall be tested in accordance with paragraph 4.6.6. Twenty-five test slabs shall be tested in accordance with 4.6.1 and 4.6.2, and thirty test slabs shall be tested in accordance with 4.6.4.

4.4.3 Acceptance/rejection criteria. The samples shall meet the examinations and tests specified by paragraphs 4.4.1 and 4.4.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.5 Inspection provisions.

4.5.1 Lotting. A lot shall consist of those faceblanks of one size produced from one lot of molding compound using one lot of latex and one lot of filter material.

4.5.1.1 Lots of molding compound and latex. A lot is defined as that quantity of material that has been manufactured at one plant by one manufacturer in one week under essentially the same manufacturing conditions.

4.5.2 Sampling.

4.5.2.1 Nondestructive examination and tests. Sampling shall be conducted in accordance with Standard MIL-STD-105. The level and degree of inspection will be as specified in the contract or order.

4.5.2.2 Destructive tests.

4.5.2.2.1 Faceblanks. From each lot, five faceblanks selected at random shall be removed for test.

4.5.2.2.2 Molding compound. Twenty-five (25) test slabs shall be prepared and tested in accordance with paragraphs 4.6.1 and 4.6.2.

4.5.3 Examination. Sample faceblanks from the regular production lots shall be examined in accordance with the classification of defects and the AQL's as designated by this specification and with Standard MIL-STD-105. A definition of terminology is presented in Standard MIL-STD-177.

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4.5.3.1 Mold approval. Prior to the initiation of production, the inspector shall ascertain that each applicable mold has been approved by the procuring activity.

4.5.4 Classification of defects.

4.5.4.1 Faceblanks.

Categories and defects

Major — AQL 1.0 percent defective

1. Incomplete or short shot
2. Sink or shrinkage marks
3. Orange peels
4. Dull streaks or flow lines
5. Weld lines
6. Blisters or bubbles
7. Burned spots
8. Excessive shrinkage around gates
9. Stock discoloration or breakdown
10. Flashing
11. Filter unit not sealed properly or incorrectly assembled
12. Latex applied incorrectly
13. Contamination (foreign matter)
14. Damage (see 3.8)

Minor — AQL 4.0 percent defective

51. Black flecks but no discoloration

4.5.4.2 Preparation for delivery. (See section 5).

Categories and defects

Major — AQL 4.0 percent defective

1. Container damaged
2. Marking incorrect, missing or illegible
3. Faceblanks deformed or damaged
4. Mixed sizes of faceblanks

Minor — AQL 10.0 percent defective

51. Non-uniform quantities packaged or packed in container

4.6 Tests. Tests shall be conducted as follows:

4.6.1 Test slabs. Test slabs selected in paragraph 4.5.2.2.2 shall be tested as specified in table II, 4.6.1.1 and 4.6.2. The average of 5 results shall be reported.

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Table II. Test methods

Property	Methods
Tensile strength	1011 Spec. L-P-406
Elongation	1011 Spec. L-P-406
Modulous	1011 Spec. L-P-406
Compression set	1021 Spec. L-P-406
Flammability	2021 Spec. L-P-406
Specific gravity	5011 Spec. L-P-406
Hardness	3021 Fed. Test Meth. Std. 601
Tear resistance	4221 Fed. Test Meth. Std. 601

4.6.1.1 Resiliant rebound.

4.6.1.1.1 Impact resiliameter. The impact resiliameter shall be the apparatus shown in figure 2. The steel ram rod shall be 35.5 ± .5 centimeters (cm.) long by 1.27 ± .02 cm. in diameter. The rod shall have a hemispherical end. The ram rod shall have no wabble when attached to the suspension panel. A flat surface steel anvil 1 inch in diameter having a device to hold the plastic specimen which is 1 inch in diameter in position is rigidly fixed so as to be centered on and almost touching the ram when the latter is at rest. The horizontal length of this scale is 62.5 cm. so that the 2 meter pendulum will drop 10 cm. vertically when swinging along the scale from 100 to 0. A pointer on the trailing end of the ram furnishes a means of releasing the ram when making a test and enables the operator to follow its travel over the scale.

4.6.1.1.2 Procedure. Place a 1 inch circle sheet of plastic into the holder, release the ram rod from the one hundred position and read the scale after it rebounds from the plastic test specimen.

4.6.1.2 Low temperature flexibility. The low temperature flexibility shall be determined in accordance with A.S.T.M. method D746.

4.6.2 Aging. Specimens for aging shall be placed into an oven for 168 hours at 158° ± 2°F. They shall be tested for the aged requirements of 3.4.1 with the applicable method of 4.6.1.

4.6.3 Faceblanks. Each faceblank sample shall be cut and tested for the requirements of paragraph 3.4.2 and 3.6. The average of 5 results shall be reported. (The hardness and specific gravity tests are in 4.6.1 and pull test in 4.6.5.)

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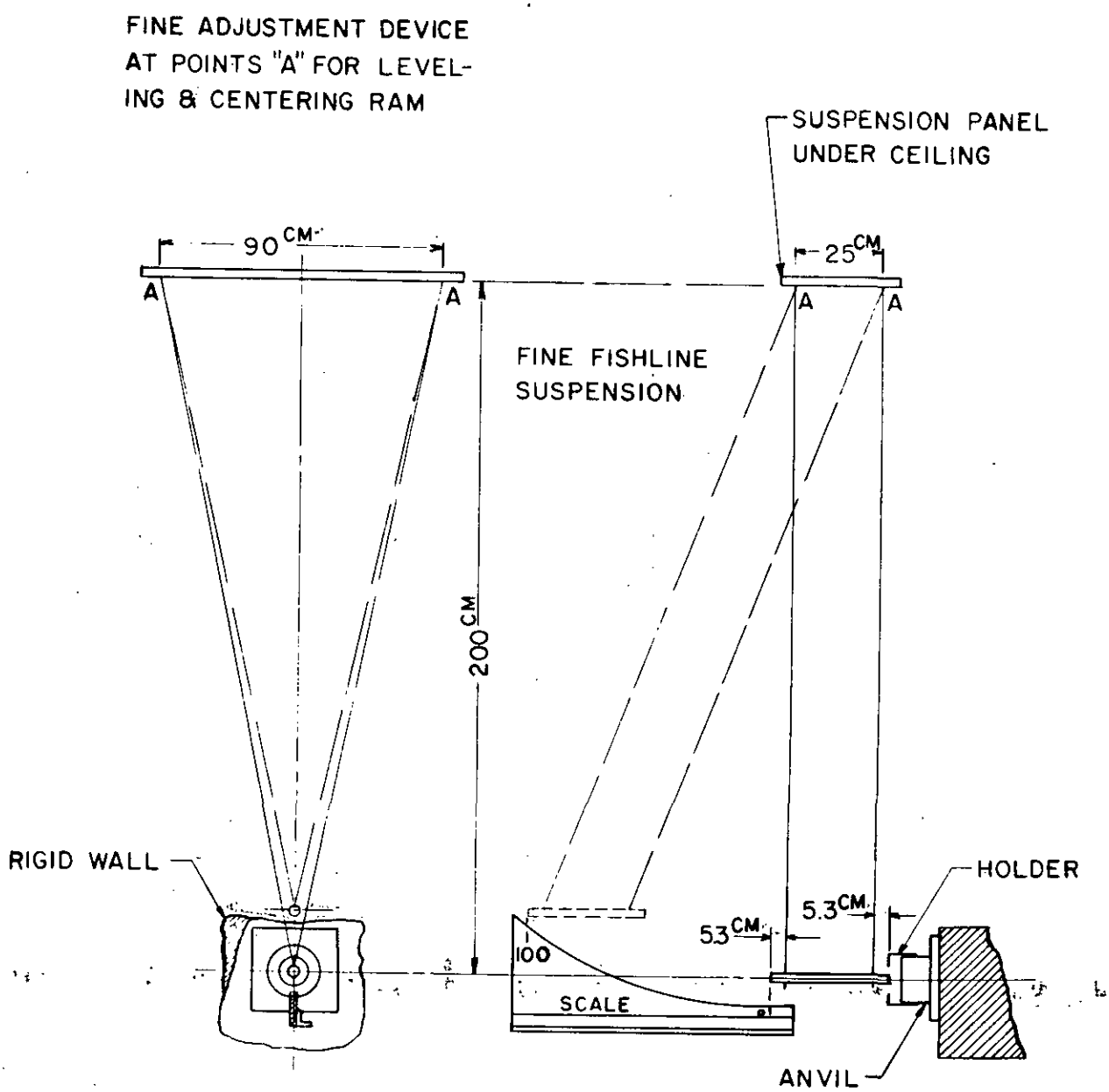


Figure 2 – Diagram of Impact Resiliometer

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4.6.4 Toxicity. All sample slabs submitted to the agency designated by the contracting officer for test will be evaluated from the toxicity standpoint by the Office of the Surgeon General, Department of the Army. Such tests, including human skin tests, as are deemed necessary by the Office of the Surgeon General for properly evaluating the toxicity of any test slab will be carried out in accordance with the directions of the Surgeon General (see 6.5).

4.6.5 Latex adhesion pull test. The pull test shall be performed in accordance with method 8311 of Fed. Test Method Std. No. 601 modified as follows: That the specimen shall be two inches long and one inch wide; that the seam shall be parallel to the width edge; that the distance between the grips be 1 ± 1/8 inches and that the rate of travel be two inches per minute.

4.6.6 Charcoal poisoning. Molded samples shall be aged at 158°F. ± 2°F. for 70 hours. They shall then be tested on Chemical Corps Apparatus, All Purpose, Gas Life Tester, Q2. The adapter shall be designed to clamp the plastic so that the whole filter element is tested. The concentration of chloropicrin and the flow rate of air shall be 5 milligrams per liter and 10 liters per minute respectively. The mean gas life shall be calculated as follows:

$$G = \frac{MA}{100} - 0.00113\ A\ \sqrt{10\ \Sigma(Y)^2 - (\Sigma Y)^2}$$

where: G = Mean gas life, in minutes.
M = Average of the determinations.
Y = Each individual determination.
A = Area of the element tested.

G must be equal to or greater than the required mean gas life as specified in Purchase Description 197-54-653 for type II material.

4.6.7 Acceptance/rejection criteria. If the test slabs or sample faceblanks fail to meet the requirements of this specification, the lot of faceblanks represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A and B. Not applicable.

5.1.2 Level C. A uniform quantity of faceblanks of one size shall be packaged in accordance with manufacturer's commercial practice, and in such a manner as to prevent any deformation or damage to the faceblank.

5.2 Packing.

5.2.1 Level A and B. Not applicable.

5.2.2 Level C. A uniform quantity of faceblanks of one size packaged as specified in 5.1.2 shall be packed in such a manner as to afford protection against any deformation or damage to the faceblanks during direct shipment from the supply source to the first receiving activity for immediate use.

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Containers used shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified by contract or order all packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. This faceblank is intended for use in the CD V-805 mask, protective, civilian.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of faceblank required.
(c) Color of faceblank.

6.3 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 Pilot lot. A pilot lot may be required to evaluate the supplier's production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

6.5 Toxicity. The technique of determining toxicity by means of patch tests is given in Schwartez, Tulipau, and Peck. "Occupational Diseases of the Skin", 2d, Ed., pages 54 to 64 and 302 to 308 (1947). This test takes from 3 to 6 months.

6.6 Latex. A latex which has been found to meet the requirements of the specification is Geon Latex No. 576 by B. F. Goodrich Co.

6.7 Supersession data. This specification supersedes Chemical Corps Purchase Description 197-54-694, dated 28 May 1959.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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